Chapter 6—Inner product spaces

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Story:

- In coming up with definition of a vector space
  - we generalised the linear structure (addition and scalar multiplication)
    of R^2 and R^3.
  - But we ignored other important features,
    - such as length and angles.
  - These ideas are embedded
    in the concept we now investigate—inner products
- In this chapter, the text covers the following
  (we only focus on what is included in 6A and 6B, so we would likely skip the last point)
  - Cauchy-Schwarz Inequality
  - Gram-Schmidt Procedure
  - Linear functionals on inner product spaces
  - Calculating minimum distance to a subspace

Notation 6.1 (F, V)

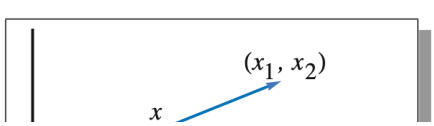
F denotes R or C.
V denotes a vector space over F.

6 A Inner Products and norms

Inner products

- Story: To motivate the concept of inner products
  think of vectors in R^2 and R^3
  as arrows that have their
  initial points at the origin.
- Defn/Notation: The length of a vector x in R^2 or R^3
  is called the **norm** of x
  and it is denoted by $||x||$.

  Thus, for $x = (x_1, x_2) \in R^2$
  we have $\big||x|\big| = \sqrt{x_1^2 + x_2^2}$.
- Defn/Notation: Similarly, if $x = (x_1, x_2, x_3) \in R^3$
  then $\big||x|\big| = \sqrt{x_1^2 + x_2^2 + x_3^2}$.



The length of this vector x is
$\sqrt{{x_1}^2 + {x_2}^2}$.

- Story: Even though we cannot draw pictures in higher dimensions
  the generalisation to R^n is obvious:
- Defn/Notation: We define the norm of $x = (x_1, \dots x_n) \in R^n$
  $$||x|| = \sqrt{x_1^2 + \cdots + x_n^2}.$$

- Remark:
  The norm is not linear in R^n
  To inject linearity into the discussion,
  one can introduce the dot product.

Definition 6.2 (dot product)

For $x, y \in R^{2n}$
the **dot product** of x and y
denoted by $x \cdot y$ is defined by
$$x \cdot y = x_1 y_1 + \cdots + x_n y_n$$

where $x = (x_1, \dots x_n)$ and $y = (y_1, \dots y_n)$.

Remark:
- Note that the *dot product* of two vectors in R^n
  is a number not a vector.

- Obviously,
  $x \cdot x = ||x||^2$ for all $x \in R^n$.

NB.
- The dot product on R^n
  has the following properties.

  - $x \cdot x \geq 0$ for all $x \in R^n$
  - $x \cdot x = 0$ iff $x = 0$
  - for $y \in R^n$ fixed
    the map from R^n to R
    that sends $x \in R^n$ to $x \cdot y$
    is linear.

  - $x \cdot y = y \cdot x$ for all $x, y \in R^n$.

Story:
- An inner product is a generalisation of the dot product
- One may be tempted to think

One may be tempted to think
an inner product is defined by
abstracting the properties of the dot product
discussed above

- For real vector spaces, the guess is correct.
- For complex, for this definition to be useful,
one needs to be a bit more careful.

- Recall that if $\lambda = a + ib$
where $a, b \in R$ then

  - The absolute value of λ, i.e. $|\lambda|$ is
defined by $|\lambda| = \sqrt{a^2 + b^2}$

  - The complex conjugate of λ
denoted by $\bar{\lambda}$
is defined by $\bar{\lambda} = a - ib$

  - $|\lambda|^2 = \lambda\bar{\lambda}$.

(the text refers the readers to Chapter 4
for a review of the basic definitions and properties of
absolute values and complex conjugation)

- For $z = (z_1 \cdots z_n) \in C^n$
we define the norm of z as follows:

$$||z|| = \sqrt{|z_1|^2 + \cdots + |z_n|^2}.$$

- The absolute values are needed because
we want $||z||$ to be a nonnegative number.

- Note that it also holds that
$||z||^2 = z_1\bar{z}_1 + \cdots z_n\bar{z}_n$.

- We want to think of $||z||^2$
as the inner product of z with itself;
as we did in R^n.

- The equation above suggests that
the inner product between $w = (w_1 \cdots w_n) \in C^n$ with z should be

$$w_1\bar{z}_1 + \cdots w_n\bar{z}_n.$$

- If the roles of w and z are interchanged
the expression above would be replaced with its
complex conjugate.

- With this motivation
we can define inner product on V

we can define inner product on V
which may be a real or a complex vector space.

Remarks (about the notation used below)

- if λ is a complex number
  then the notation $\lambda \geq 0$ means
  λ is real and non-negative.

- We use
  $\langle u, v \rangle$ to denote an inner product
  Some people use parenthesis (u, v)
  but here it may get ambiguous
  it could denote either an ordered pair or
  an inner product.

Definition 6.3 (inner product)

An inner product on V
is a function that takes
each ordered pair (u, v)
of elements of V
to
a number $\langle u, v \rangle \in F$
and has the following properties:

Positivity
$\langle v, v \rangle \geq 0$ for all $v \in V$.

Definiteness
$\langle v, v \rangle = 0$ iff $v = 0$.

additivity in first slot
$\langle u + v, w \rangle = \langle u, w \rangle + \langle v, w \rangle$ for all $u, v, w \in V$

homogeneity in first slot
$\langle \lambda u, v \rangle = \lambda \langle u, v \rangle$ for all $\lambda \in F$ and all $u, v \in V$.

conjugate symmetry
$\langle u, v \rangle = \overline{\langle v, u \rangle}$ for all $u, v \in V$.

Remark about conventions:
Most mathematicians define inner product as above
many physicists use a definition that
requires homogeneity in the second slot
instead of the first slot.

Remark (restricting to reals):

- Every real number equals its complex conjugate.
- Thus, if we are dealing with a real vector space
  the last condition can be simplified to
  $\langle u, v \rangle = \langle v, u \rangle$.

Example 6.4 (inner products)

(a) The Euclidean inner product on F^n
is defined by
$$\langle (w_1, \dots w_n), (z_1, \dots z_n) \rangle = w_1 \bar{z}_1 + \cdots + w_n \bar{z}_n$$

(b) If $c_1, \dots c_n$ are positive numbers
then an inner product can be defined on F^n by

$$\langle (w_1, \dots w_n), (z_1, \dots z_n) \rangle = c_1 w_1 \bar{z}_1 + \cdots + c_n w_n \bar{z}_n.$$

(c) An inner product can be defined on the vector space of
continuous real-valued functions on the interval
$[-1,1]$
$$\langle f, g \rangle = \int_{-1}^{1} f(x) g(x) dx.$$

(d) An inner product can be defined on $\mathcal{P}(R)$ by
$$\langle p, q \rangle = \int_0^\infty p(x) q(x) e^{-x} dx.$$

Definition 6.5 (inner product space)
An inner product space
is a vector space V along an inner product V.

Story:
- The most important example of an inner product space
  is F^n with the Euclidean inner product
  given by part (a) of the last example.

Convention:
- When F^n is referred to as an inner product spcae
  assume that the inner product is the Eucledian inner product
  unless explicitly told otherwise.

Story:
- To not have to repeat that V is an inner product space,
  for the remainder of this chapter
  we make the following assumption.

Notation 6.6 (V)
For the rest of this chapter, V denotes an inner product space over F.

Story:
- We abused the notation slightly here.
- An inner product space
  is a vector space, along with an inner product on that vector space
- When we say
  a vector space V is an inner product space
  we are thinking that an inner product on V

is clear from the context
(e.g. for F^n, the inner product will be taken to be the Euclidean inner product)

Proposition 6.7 (Basic properties of an inner product)

(a) For each fixed $u \in V$
the function that takes v to $\langle v, u \rangle$
is a linear map from V to F.

(b) $\langle 0, u \rangle = 0$ for every $u \in V$.

(c) $\langle u, 0 \rangle = 0$ for every $u \in V$.

(d) $\langle u, v + w \rangle = \langle u, v \rangle + \langle u, w \rangle$ for all $u, v, w \in V$.

(e) $\langle u, \lambda v \rangle = \bar{\lambda} \langle u, v \rangle$ for all $\lambda \in F$ and $u, v \in V$.

Proof.

(a)
We basically need to show that
$\langle v + aw, u \rangle = \langle v, u \rangle + a\langle w, u \rangle$

and this follows from
the additivity property and homogeneity property
in the first slot

Positivity
$\langle v, v \rangle \geq 0$ for all $v \in V$.

Definiteness
$\langle v, v \rangle = 0$ iff $v = 0$.

additivity in first slot
$\langle u + v, w \rangle = \langle u, w \rangle + \langle v, w \rangle$ for all $u, v, w \in V$

homogeneity in first slot
$\langle \lambda u, v \rangle = \lambda \langle u, v \rangle$ for all $\lambda \in F$ and all $u, v \in V$.

conjugate symmetry
$\langle u, v \rangle = \overline{\langle v, u \rangle}$ for all $u, v \in V$.

(b) This follows from part (a): a linear map must map 0 to 0.

(c) This, in turn, follows from the conjugate symmetry property
$\langle u, 0 \rangle = \overline{\langle 0, u \rangle} = 0$

(d) Suppose $u, v, w \in V$.
Then

$$\begin{aligned}
\langle u, v + w \rangle &= \overline{\langle v + w, u \rangle} \\
&= \overline{\langle v, u \rangle + \langle w, u \rangle} \\
&= \overline{\langle v, u \rangle} + \overline{\langle w, u \rangle} \\
&= \langle u, v \rangle + \langle u, w \rangle.
\end{aligned}$$

(e) Basically follows from the conjugate symmetry property:
Suppose $\lambda \in F$ and $u, v \in V$. Then

$$\begin{aligned}
\langle u, \lambda v \rangle &= \overline{\langle \lambda v, u \rangle} \\
&= \overline{\lambda \langle v, u \rangle} \\
&= \bar{\lambda} \overline{\langle v, u \rangle} \\
&= \bar{\lambda} \langle u, v \rangle,
\end{aligned}$$

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Norms

Story:

Our motivation for defining inner products
came initially from the norms of vectors
on R^2 and R^3.

Now, we use inner products to define a norm.

Definition 6.8 (norm, $||v||$)

For $v \in V$,
the **norm** of v
denoted by $||v||$ is defined by
$$\big|\big|v\big|\big| = \sqrt{\langle v, v\rangle}.$$

Example 6.9 (norms)

(a)

If $(z_1, \ldots, z_n) \in \mathbf{F}^n$ (with the Euclidean inner product), then
$$\|(z_1, \ldots, z_n)\| = \sqrt{|z_1|^2 + \cdots + |z_n|^2}.$$

(b)

In the vector space of continuous real-valued functions on $[-1, 1]$ [with inner product given as in part (c) of 6.4], we have
$$\|f\| = \sqrt{\int_{-1}^{1} \big(f(x)\big)^2\, dx}.$$

Proposition 6.10 (basic properties of the norm)

Suppose $v \in V$.

(a) $\big|\big|v\big|\big| = 0$ iff $v = 0$.
(b) $\big|\big|\lambda v\big|\big| = |\lambda| \cdot ||v||$ for all $\lambda \in F$.

Positivity
$\langle v, v\rangle \geq 0$ for all $v \in V$.

Definiteness
$\langle v, v\rangle = 0$ iff $v = 0$.

additivity in first slot
$\langle u + v, w\rangle = \langle u, w\rangle + \langle v, w\rangle$ for all $u, v, w \in V$

homogeneity in first slot
$\langle \lambda u, v\rangle = \lambda\langle u, v\rangle$ for all $\lambda \in F$ and all $u, v \in V$.

conjugate symmetry

Proof.

(a) This is because $\langle v, v\rangle = 0$ iff $v = 0$ (from the "definiteness property).
(b) Suppose $\lambda \in F$.
Then
$$\begin{aligned}\|\lambda v\|^2 &= \langle \lambda v, \lambda v\rangle \\ &= \lambda\langle v, \lambda v\rangle \\ &= \lambda\bar{\lambda}\langle v, v\rangle\end{aligned}$$

$$= |\lambda|^2 \|v\|^2.$$

Take square roots.

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conjugate symmetry

$\langle u, v\rangle = \overline{\langle v, u\rangle}$ for all $u, v \in V$.

Definition 6.11 (orthogonal)

Two vectors $u, v \in V$ are called orthogonal

if $\langle u, v\rangle = 0$.

Story:

- We begin our study of orthogonality with an easy result.

Proposition 6.12 (Orthogonality and 0)

(a) 0 is orthogonal to every vector in V.

(b) 0 is the only vector in V that is orthogonal to itself.

Proof.

(a) Part (b) of Prop 6.7 states that

$\langle 0, u\rangle = 0$ for every $u \in V$.

(b) If $v \in V$ and $\langle v, v\rangle = 0$

then $v = 0$ (by definition of the inner product).

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Proposition 6.7 (Basic properties of an inner product)

(a) For each fixed $u \in V$

the function that takes v to $\langle v, u\rangle$

is a linear map from V to F.

(b) $\langle 0, u\rangle = 0$ for every $u \in V$.

(c) $\langle u, 0\rangle = 0$ for every $u \in V$.

(d) $\langle u, v + w\rangle = \langle u, v\rangle + \langle u, w\rangle$ for all $u, v, w \in V$.

(e) $\langle u, \lambda v\rangle = \bar{\lambda}\,\langle u, v\rangle$ for all $\lambda \in F$ and $u, v \in V$.

For the special case $V = R^2$

the next theorem has been known for over 2,500 years.

Of course

the proof below is not the original one.

Theorem 6.13 (Pythagorean Theorem)

Suppose u and v are orthogonal vectors in V.

Then

$$\left|\left|u + v\right|\right|^2 = \left|\left|u\right|\right|^2 + \left|\left|v\right|\right|^2.$$

Proof.

We have

$$\begin{aligned}\|u + v\|^2 &= \langle u + v, u + v\rangle \\ &= \langle u, u\rangle + \langle u, v\rangle + \langle v, u\rangle + \langle v, v\rangle \\ &= \|u\|^2 + \|v\|^2,\end{aligned}$$

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Story:

....

<skipped the discussion between this and Cauchy-Schwarz>

Proposition 6.14 (an orthogonal decomposition)

Suppose $u, v \in V$ with $v \neq 0$.
Set $c = \langle u, v \rangle$ and $w = u - \frac{\langle u,v \rangle}{||v||^2}$.

Then
$\langle w, v \rangle = 0$ and $u = cv + w$.

Proposition 6.15 (Cauchy-Schwarz Inequality)
Suppose $u, v \in V$.
Then
$|\langle u, v \rangle| \leq ||u|| \cdot ||v||$.

This inequality is an equality
iff
u is a scalar multiple of v (or, equivalently, the other way).

Proof If $v = 0$, then both sides of the desired inequality equal 0. Thus we can assume that $v \neq 0$. Consider the orthogonal decomposition

$$u = \frac{\langle u, v \rangle}{\|v\|^2} v + w$$

given by 6.14, where w is orthogonal to v. By the Pythagorean Theorem,

$$\begin{aligned} \|u\|^2 &= \left\| \frac{\langle u, v \rangle}{\|v\|^2} v \right\|^2 + \|w\|^2 \\ &= \frac{|\langle u, v \rangle|^2}{\|v\|^2} + \|w\|^2 \\ &\geq \frac{|\langle u, v \rangle|^2}{\|v\|^2}. \end{aligned} \tag{6.16}$$

Multiplying both sides of this inequality by $\|v\|^2$ and then taking square roots gives the desired inequality.

Looking at the proof in the paragraph above, note that the Cauchy–Schwarz Inequality is an equality if and only if 6.16 is an equality. Obviously this happens if and only if $w = 0$. But $w = 0$ if and only if u is a multiple of v (see 6.14). Thus the Cauchy–Schwarz Inequality is an equality if and only if u is a scalar multiple of v or v is a scalar multiple of u (or both; the phrasing has been chosen to cover cases in which either u or v equals 0). ■